

SOFTWARE ENGINEERING LAB 1

EV Charging Station Reservation System

Problem Statement #21

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SECTION : C

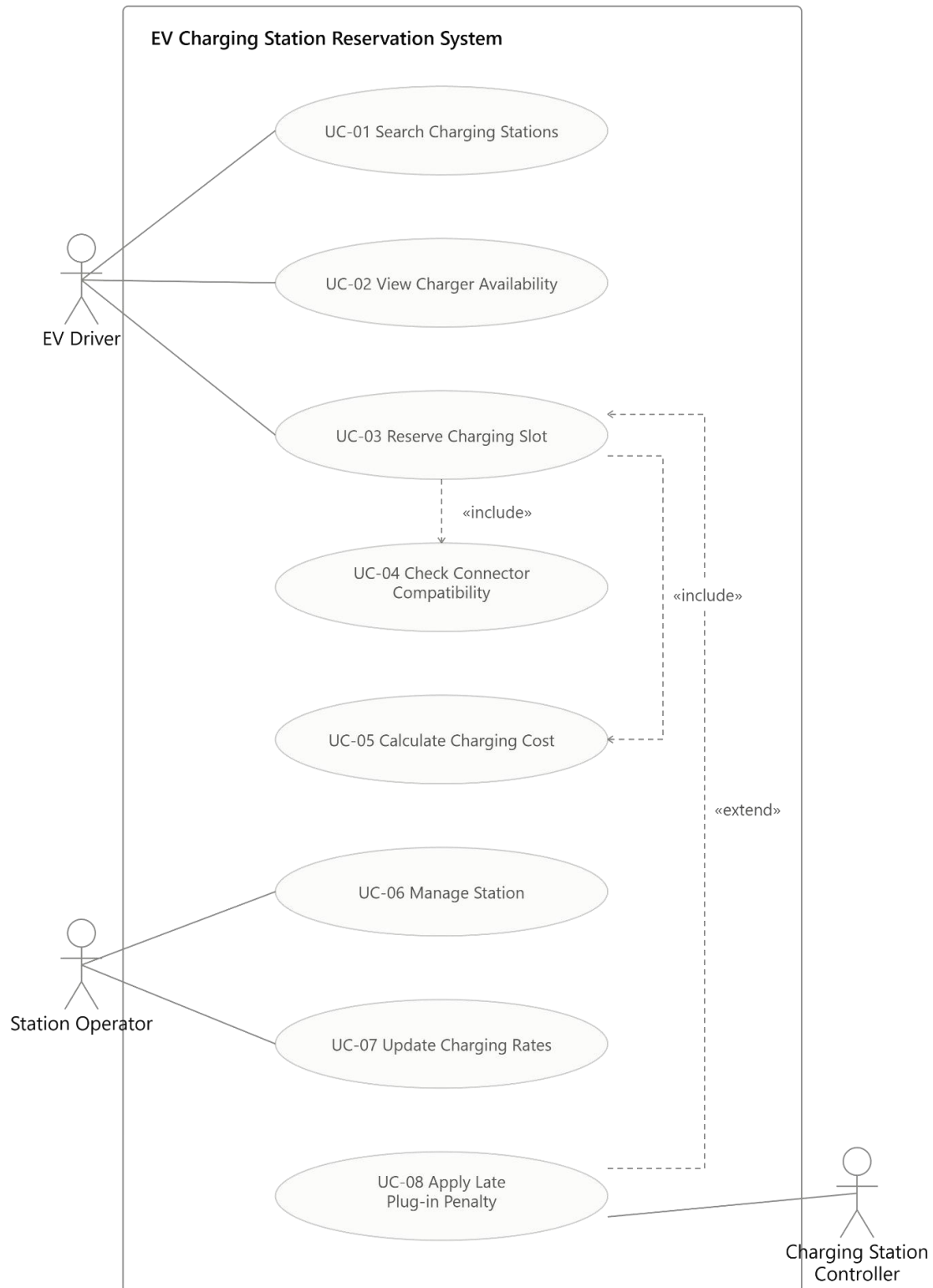
1. Requirements Table

Functional and nonfunctional requirements for the EV Charging Station Reservation System.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match the driver's vehicle connector type with available station chargers and allow the driver to reserve a 30-minute charging window.	High	Pass: Only compatible chargers can be selected and a 30-minute slot is reserved. Fail: an incompatible charger can be selected or the same socket is booked by two vehicles.	Main function needed to give the driver a suitable charger and time slot.
FR-002	Functional	The system shall allow an EV driver to locate charging stations using a location.	High	Pass: Charging stations available for the entered location are displayed. Fail: registered stations for that location are not shown.	Drivers need to find a suitable charging station.
FR-003	Functional	The system shall show the available charging sockets and time slots for a selected station.	High	Pass: Available sockets and time slots are displayed. Fail: a reserved socket or slot is shown as available.	The driver needs availability information before making a reservation.
FR-004	Functional	The system shall calculate the charging cost using the current dynamic power rate.	High	Pass: The displayed cost uses the current power rate. Fail: the displayed rate is missing or outdated.	Dynamic power-rate calculation is part of the given problem.
FR-005	Functional	The system shall allow the station operator to update charger availability, connector information, and charging rates.	Medium	Pass: An operator update is reflected in the station information. Fail: the changed information is not updated.	The operator needs to keep station information accurate.
NFR-001	Nonfunctional – Performance & Security	The system shall automatically release a locked socket and apply the applicable penalty when the driver does not plug in within 10 minutes of the slot start.	High	Pass: After 10 minutes without plug-in, the socket is released and the penalty is recorded. Fail: either action is missing.	This follows the supplied requirement and prevents unused reservations from blocking sockets.
NFR-002	Nonfunctional – Reliability	The system shall prevent two drivers from successfully reserving the same socket for overlapping time slots.	High	Pass: Concurrent booking tests result in only one successful reservation. Fail: two reservations are confirmed.	Reliable booking is necessary to avoid double reservations.

2. UML Use-Case Diagram

EV Charging Station Reservation System – Use-Case Diagram



3. Use-Case Flow

Use Case: UC-03 – Reserve Charging Slot

Primary Actor: EV Driver

Goal: Reserve a compatible charging socket for a 30-minute charging window.

Preconditions

- The EV driver can access the reservation system.
- The selected charging station is available in the system.
- The driver has a vehicle connector type.
- A compatible charger and a free time slot are available.

Postconditions

- The selected 30-minute slot is reserved.
- The selected socket is unavailable for overlapping reservations.
- The applicable dynamic power rate is used for the charging cost.
- The driver receives confirmation.

Main Success Scenario

1. The EV Driver searches for charging stations using a location.
2. The system displays matching stations and compatible charger information.
3. The EV Driver selects a charging station.
4. The system displays available sockets and time slots.
5. The EV Driver selects a 30-minute time slot.
6. The system checks the vehicle connector against the selected charger.
7. The system calculates the charging cost using the current dynamic power rate.
8. The system displays the reservation details and estimated cost.
9. The EV Driver confirms the reservation.
10. The system reserves the selected socket and time slot.
11. The system shows the reservation confirmation.

Alternate Flow – Incompatible Connector

12. At Step 6, the system finds that the selected charger is not compatible with the driver's connector.
13. The system does not allow the reservation for that charger.
14. The EV Driver selects another compatible charger or slot.
15. The system continues the reservation process.